

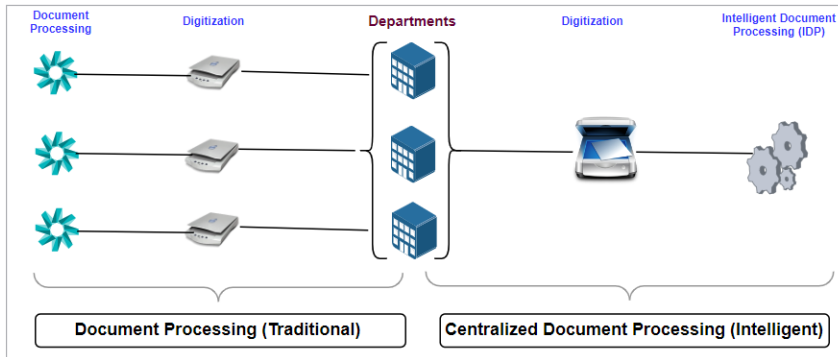


Figure 3: The evolution of document processing

extraction. An intuitive UI will help the labellers tasked with this responsibility. Every machine learning project needs to factor in estimates to accommodate a percentage split between straight-thru processing and HITL.

Overall, IDP has enhanced accuracy levels, making document processing more seamless without compromising on scale. However, for IDP to work optimally, more sample data and associated model training are necessary. Accessing sample data may not be easy and straightforward, but this can be compensated to some extent by relying on synthetic data.

How LLMs augment the capabilities of IDP

Large language models (LLMs) help advance document processing as they can simply work by providing the right prompts without explicit training. There is no need for sample data as they are pre-trained on the enterprise corpus of data. However, they are more suitable for key-value (KV) scenarios. When there is a combination of KV pairs along with complex unstructured content, a composite model that blends traditional ML models with LLMs can be effective.

In spite of all this, there is still a chance of hallucinations. The latest developments in the form of RAG (retrieval-augmented generation) have introduced a next level of validation through external validation(s), amplifying the LLM output by providing a better overall response.

How to choose the right IDP platform

With numerous choices in the market for IDP, choosing the right one can be challenging. The following criteria/features can help in making a selection.

- Intuitive low code interface/platform
- Flexibility to choose vendor-agnostic OCR engine
- Ability to have vendor-agnostic cloud service provider
- Option to create/configure document taxonomies
- Multiple language support
- Support for auto ML models
- Augment ML models with LLMs and RAG features
- Ability to continuously up-train for better accuracy
- Option for exception handling through human operators in the loop
- Seamless integrations and connectors
- Ability to create complex workflows allowing effortless collaborations
- Processing of large document batches
- Support for retention policies/rules

- Dashboards and reporting

IDP value proposition

Documents are pivotal for any business function, cutting across domains like finance/banking, healthcare, public sector, insurance, retail, etc. Important use cases include mortgage, invoice processing, claims, KYC (know your customer), EMR/EHR, etc.

Traditionally, document processing is more siloed at the department level. To derive better results and have cross-pollination of best practices, it is recommended to converge this into a centralised document processing shared service. IDP comes in handy to promote this by sharing models and best practices, enabling reusability across the organisation.

Overall, it is not a one-size-fits-all kind of setup. Document processing can be done based on factors like complexity, volume, etc. If the content is structured with a standard format/layout, then an OCR-based solution should be sufficient. However, if the content is complex, mapped to different document types, with a need for extraction of hand-written text along with a greater emphasis on accuracy, then AI-ML based intelligent document processing makes a perfect fit.

To ensure the most informed decision, assess and evaluate an organisation's document processing needs. Consider factors like budget, timelines, and desired accuracy levels to arrive at the better option suitable for the organisation. **END** 🐧

Reference

- <https://www.globenewswire.com/news-release/2023/11/21/2783889/0/en/Intelligent-Document-Processing-Market-Size-to-Surpass-USD-12-81-billion-by-2030-exhibiting-a-CAGR-of-32-9.html>

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